

Homework: Differential and Difference Equations

GSE 510

Due: Next Class

Instructions: Answer all questions carefully. Show your work. The goal of this assignment is to practice solving simple first-order differential equations and difference equations, and to interpret them in economic settings.

Practice Problems: Work together in class

1. Consider the difference equation

$$x_{t+1} = 0.8x_t + 10.$$

- (a) Find the steady state x^* .
- (b) Suppose $x_0 = 20$. Compute x_1 , x_2 , and x_3 .
- (c) Is the sequence moving toward the steady state?
- (d) Is the steady state stable? Explain.
- (e) Give an economic interpretation of this equation if x_t is capital.

2. Consider the difference equation

$$y_{t+1} = 1.2y_t.$$

- (a) Find the steady state.
- (b) Suppose $y_0 = 5$. Compute y_1 , y_2 , and y_3 .
- (c) Does the sequence converge or diverge?
- (d) Is the steady state stable?
- (e) Explain the role of the coefficient 1.2.

3. Consider the first-order differential equation

$$\frac{dx}{dt} = 6 - 2x.$$

- (a) Find the steady state x^* .
- (b) If $x < x^*$, is x increasing or decreasing?
- (c) If $x > x^*$, is x increasing or decreasing?
- (d) Is the steady state stable?

4. Consider the differential equation

$$\frac{dk}{dt} = sAk - \delta k.$$

Assume

$$s = 0.2, \quad A = 4, \quad \delta = 0.5.$$

- (a) Substitute the parameter values into the equation.

- (b) Simplify the equation.
- (c) Does $k(t)$ grow or shrink over time if $k(0) > 0$?
- (d) Find the growth rate of $k(t)$.
- (e) Interpret this equation as a simple capital accumulation equation.

5. Consider the difference equation

$$p_{t+1} = p_t + \alpha(D(p_t) - S(p_t)),$$

where

$$D(p) = 100 - 2p, \quad S(p) = 20 + p, \quad \alpha = 0.1.$$

- (a) Substitute $D(p)$, $S(p)$, and α into the equation.
- (b) Simplify the difference equation.
- (c) Find the steady-state price p^* .
- (d) Is the steady state stable?
- (e) Interpret the adjustment rule economically.

Homework Problems: Submit individually

6. Consider the difference equation

$$x_{t+1} = 0.5x_t + 12.$$

- (a) Find the steady state x^* .
- (b) Suppose $x_0 = 0$. Compute x_1 , x_2 , and x_3 .
- (c) Is the sequence converging to the steady state?
- (d) Write the equation in the form

$$x_{t+1} - x^* = a(x_t - x^*).$$

- (e) Use part (d) to explain why the steady state is stable.

7. Consider the difference equation

$$y_{t+1} = 1.1y_t - 5.$$

- (a) Find the steady state y^* .
- (b) Suppose $y_0 = 60$. Compute y_1 , y_2 , and y_3 .
- (c) Does the sequence move toward or away from the steady state?
- (d) Is the steady state stable?
- (e) Explain why the coefficient on y_t matters for stability.

8. Consider the differential equation

$$\frac{dx}{dt} = 4x - 20.$$

- (a) Find the steady state.
- (b) If $x > x^*$, is x increasing or decreasing?
- (c) If $x < x^*$, is x increasing or decreasing?
- (d) Is the steady state stable or unstable?

9. Suppose price adjusts according to

$$\frac{dp}{dt} = \gamma(D(p) - S(p)),$$

where

$$D(p) = 80 - p, \quad S(p) = 20 + 2p, \quad \gamma = 0.5.$$

- (a) Substitute the demand, supply, and adjustment parameter into the differential equation.
- (b) Simplify the differential equation.
- (c) Find the steady-state price.
- (d) Is the steady-state price stable?
- (e) Interpret the adjustment process economically.

10. **Slightly harder problem.** Consider the Solow-style capital accumulation equation

$$k_{t+1} = sAk_t^{1/2} + (1 - \delta)k_t,$$

where

$$s = 0.4, \quad A = 10, \quad \delta = 0.2.$$

- (a) Substitute the parameter values into the equation.
- (b) Find the steady state k^* , where $k_{t+1} = k_t = k^*$.
- (c) Suppose $k_0 = 100$. Compute k_1 .
- (d) Suppose $k_0 = 900$. Compute k_1 .
- (e) Based on parts (c) and (d), explain whether capital moves toward the steady state.
- (f) Give an economic interpretation of the steady state.

Reminder: Difference Equations

A first-order linear difference equation often has the form

$$x_{t+1} = ax_t + b.$$

A steady state x^* satisfies

$$x^* = ax^* + b.$$

If $a \neq 1$, then

$$x^* = \frac{b}{1-a}.$$

The steady state is stable when

$$|a| < 1.$$

Reminder: Differential Equations

A simple first-order differential equation often has the form

$$\frac{dx}{dt} = f(x).$$

A steady state x^* satisfies

$$f(x^*) = 0.$$

To study stability, check the sign of $f(x)$ to the left and right of x^* :

- If arrows point toward x^* , the steady state is stable.
- If arrows point away from x^* , the steady state is unstable.